

Imaging the Brain

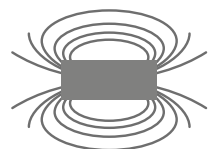
Modern medicine and neuroscience can see through the skull to observe structures within the living brain. However, imaging this soft and intricate organ has required the invention of advanced technology.

MRI scanners

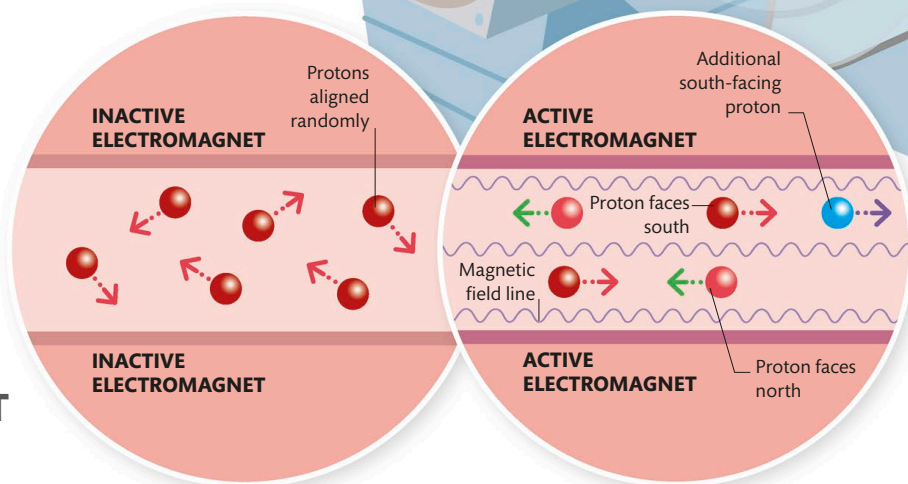
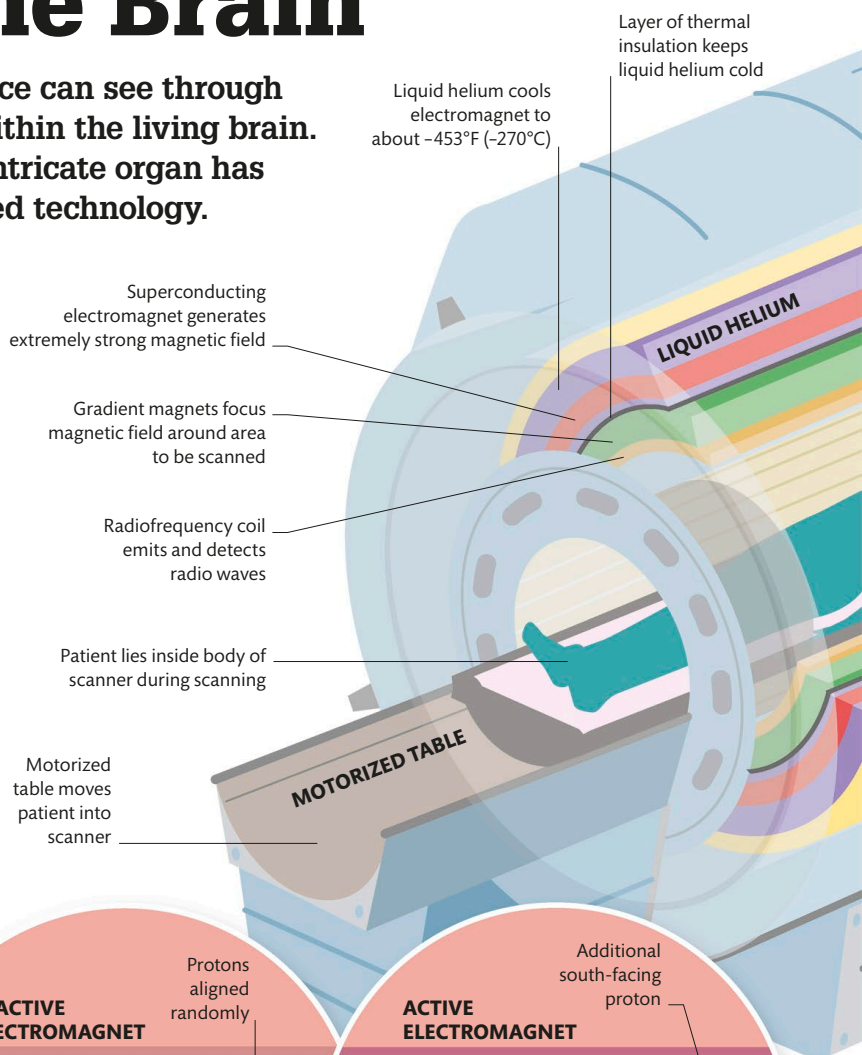
A magnetic resonance imaging (MRI) machine gives the best general view of the brain's nervous tissue and is most often deployed to search for tumors. MRI does not expose the brain to high-energy radiation, unlike other scanning systems, which makes it safe to use for long periods and multiples times. Two refinements of MRI, called fMRI and DTI, are also useful for monitoring brain activity (see p.43). Although ideal as a tool for research and diagnosis, MRI is expensive. With its liquid-helium coolant system and superconducting electromagnets, one machine also uses the power of six family homes.

How MRI works

MRI makes use of the way that protons in hydrogen atoms align to magnetic fields. Hydrogen is found in water and fats, which are both common in the brain. A scan takes about an hour, then the data is processed to create detailed images.



**THE ELECTROMAGNET
IN AN MRI SCANNER
CAN GENERATE A
MAGNETIC FIELD
40,000 TIMES AS
STRONG AS EARTH'S**



1 Protons unaligned
Before the MRI machine is activated, the protons in the brain's molecules are unaligned—the axes around which the particles are spinning point in random directions.

2 Protons align to magnetic field
Activating the machine's powerful magnetic field forces all the protons to align with each other. Approximately half face the field's north pole, and half face south. However, one pole will always have slightly more protons facing it than the other.